Fiscal Year:	FY 2004	**Task Last Updated:**	FY 01/20/2005
PI Name:	Wolgemuth, Debra J. Ph.D.		
Project Title:	Development and Function of the Mouse Vestibular System in the Absence of Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	10032	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	07/15/1999	**End Date:**	07/14/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Murashov, Alexander (East Carolina State University)		
Grant/Contract No.:	NAG2-1345		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The hypothesis tested in this research is that the absence of gravity perception, such as would occur in space, affects the development and function of the vestibular and central nervous systems. Further, we postulated that these effects will be more significant at specific stages of post-natal development of the animal. We also proposed the use of molecular genetic approaches that will provide important information as to the hierarchy of gene function during the development and subsequent function of the vestibular system. The tilted (tlt) mutant mouse has been characterized as lacking the ability to provide sensory input to the gravity receptors. The tlt/tlt mutant mice are a particularly attractive model for the study of vestibular function since the primary defect is limited to the receptor part of the vestibular system, and there are no detectable abnormal phenotypes in other organ systems. The goal of the studies was to assess immediate and delayed effects of the lack of gravity perception on the vestibular system. Particular attention was paid to characterizing primarily affected periods of vestibular morphogenesis, and to identifying downstream genetic pathways that are altered in the CNS of the tlt/tlt mutant mouse. The specific aims were: (1) to characterize the postnatal morphogenesis of the		

	CNS in the tlt mutant mouse, using detailed morphometric analysis of isolated vestibular ganglia and brain tissue at different stages of postnatal development and assessment of apoptotic cell death; (2) to examine the expression of selected genes implicated by mutational analysis to be important in vestibular development or function by in situ hybridization or immunohistochemistry in the mutant mice; and (3) to identify other genes involved in vestibular development and function, using differential cloning strategies to isolate genes whose expression is changed in the mutant versus normal vestibular system.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Understanding the genetic and molecular basis of postnatal development of the sensory systems will provide important insights for the development of new approaches for correction and treatment of sensory deficits. At present, little is known about the molecular mechanism(s) by which sensory deprivation and vestibular deprivation in particular exerts its effect on neural development. Part of our analysis was to identify those genes whose expression patterns are affected as a consequence of postnatal vestibular sensory deprivation, specifically the ability to perceive the gravity stimulus.
Task Progress:	This final year represented a no-cost extension of the original 3-year grant; hence, no new experiments were performed. We had previously submitted a manuscript which was published last year (Smith et al. Dev. Brain Res. 140:133-135, 2003). We have some additional data on apoptosis in the developing vestibular system (during embryonic development) which we hope to tally and submit for publication. The colony of het and tlt mice has now been depleted; however, mice of these genotypes can be obtained from The Jackson Lab.
Bibliography Type:	Description: (Last Updated: 02/27/2004)